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Evaluating the effect of detail in prompting Large Language Models in generating dialogue for simulation exercises

--Manuscript Draft--

Manuscript Number:	JGME-D-26-00785
Full Title:	Evaluating the effect of detail in prompting Large Language Models in generating dialogue for simulation exercises
Short Title:	Prompting details to generate dialogue
Article Type:	Original Research
Keywords:	Large Language Models; Dialogue Generation; Postgraduate Medical Education
Abstract:	Background: Prior work on the use of Large Language Models (LLMs) in medical education has shown that standardized prompts can generate accurate dialogue in patient and physician interactions. Objective: To evaluate the efficacy of using LLMs to generate dialogue for simulation exercises in emergency pediatrics, specifically with detailed prompting. Methods: Study was done in 2026. We prompted 4 LLMs with two scenarios relevant to pediatric acute care. For each condition, we prompted each LLM in two different ways; one where we used a standardized prompt, and another with a detailed flow chart instructing the LLM how the scenario should play out. Computational text analysis was conducted on length of scripts by way of prompting condition, and 3 expert clinicians from our team evaluated the usability and accuracy of the dialogues. Results: Descriptive results show that on average, outputs by model or prompt condition do not meet the cut off for high-quality responses as determined by prior research. Results from a Poisson regression model revealed that detailed flow chart prompts generated shorter dialogues compared to standardized prompts. Evaluation of each model's performance across rater, scenario, and prompt condition showed no significant difference in overall quality. Conclusions: Our findings should guide clinicians and educators in choosing LLM and prompting strategies for generating educational content as well as research on evaluating LLM outputs.

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Dear Dr. Gail M. Sullivan

Please consider our article “Evaluating the effect of detail in prompting Large Language Models in generating dialogue for simulation exercises” for publication in the *Journal of Graduate Medical Education*. This article is co-authored with Devin L Johnson, Matthew Sibbald, Quang Ngo, and Jeffrey McCarthy.

In our paper we apply previously used evaluation methods on Large Language Models (LLMs) to examine how effective they are at generating dialogue in simulation exercises within emergency pediatrics. Prior work has suggested LLMs are adept at providing high-quality dialogues sufficient for learning exercises for physician/patient interactions in within cardiac surgery procedures. We extend on this by examining if it is possible to yield similar quality dialogues for emergency pediatric procedures. Our results suggest the emergency pediatric context does not yield similar high-quality dialogues as per similar evaluation metrics. In addition, we find no effect of change in quality of dialogues, by expert raters, when we adjust the style of prompting to LLMs; one condition wherein we provide the LLM with a standard text-based prompt and another wherein we supply the LLMs with a detailed flow chart providing insight into the correct procedures to address each scenario.

This manuscript is not concurrently submitted for publication to other journals and will not be submitted elsewhere until a final decision is made by the *Journal of Graduate Medical Education*. This project is supported in part by funding from the PSI foundation. The authors declare no other potential conflicts of interest.

Thank you for your consideration.

Sincerely,

Elif Bilgic, PhD. on behalf of the co-authors.

Title: Evaluating the effect of detail in prompting Large Language Models in generating dialogue for simulation exercises

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Abstract

Background: Prior work on the use of Large Language Models (LLMs) in medical education has shown that standardized prompts can generate accurate dialogue in patient and physician interactions.

Objective: To evaluate the efficacy of using LLMs to generate dialogue for simulation exercises in emergency pediatrics, specifically with detailed prompting.

Methods: Study was done in 2026. We prompted 4 LLMs with two scenarios relevant to pediatric acute care. For each condition, we prompted each LLM in two different ways; one where we used a standardized prompt, and another with a detailed flow chart instructing the LLM how the scenario should play out. Computational text analysis was conducted on length of scripts by way of prompting condition, and 3 expert clinicians from our team evaluated the usability and accuracy of the dialogues.

Results: Descriptive results show that on average, outputs by model or prompt condition do not meet the cut off for high-quality responses as determined by prior research. Results from a Poisson regression model revealed that detailed flow chart prompts generated shorter dialogues compared to standardized prompts. Evaluation of each model's performance across rater, scenario, and prompt condition showed no significant difference in overall quality.

Conclusions: Our findings should guide clinicians and educators in choosing LLM and prompting strategies for generating educational content as well as research on evaluating LLM outputs.

Introduction

Simulation in medical education offers learners opportunities to not only develop skills for common procedures, but also high-stakes ones that are rare for learners to encounter during clinical rotations [1]. One area of simulation showcasing promise is the ability to present learners with dialogues, including between physicians/learners and patients, caregivers, or allied health team members. Through these simulated dialogues, learners could experience examples of interactions (both done well and interactions that need improvement). Additionally, learners could then have a facilitated debrief session to explore what went well or what could have been done differently to improve the care interactions. However, such approaches require considerable time and effort in drafting lengthy dialogues for such scenarios.

Artificial intelligence, specifically large language models (LLMs), have demonstrated usefulness in generating learning materials for medical education purposes [2-4]. Standardized prompts can be used to generate materials for training in clinical scenarios, allowing learners to apply general skills to specific examples [2-3]. Specifically, recent evidence suggests that LLMs can also be used to generate mock dialogues to support learner training, which can provide a base for the simulation exercises. The potential for AI generated dialogue could ease one aspect of the burden in the creation of such materials.

In a recent study, researchers used 4 popular LLMs (ChatGPT, Claude, Copilot, and Gemini) to simulate dialogues between patient and physician for 10 different cardiac procedures [5]. The dialogues, focused on both pre and post procedural interactions, were generated using zero shot prompting, a standardized single text-based prompt. Results found no differences in models in terms of overall accuracy and quality. To extend on this work, we investigate whether providing structured guidance by way of a flow chart improves the accuracy and usability of LLM generated dialogue in high-stakes pediatric scenarios compared to standardized text based prompts.

Methods

In an evaluation study done in 2026, we prompted 4 popular LLM models; ChatGPT, Claude, Copilot, and DeepSeek, to generate dialogue for two simulation scenarios in pediatrics. We originally aimed to use Google Gemini, as this was one of the LLMs used in our reference study [5]; however, during the prompting for our scenarios, we were unable to generate dialogues with Gemini with the model citing safety concerns and suggesting alternative topics to prompt with. We thus replaced Gemini with the DeepSeek model as it has been used in similar studies evaluating dialogue material [1]. We used free versions of each model and all dialogues generated came from a new account created for the purposes of this research, so no prior information such as chat history could be inferred by way of each LLM.

Each LLM generated dialogues for two scenarios relevant to pediatric acute care. Both scenarios were selected as representative of high acuity, low-occurrence events commonly encountered in emergency medical contexts. The first scenario was about pediatric cardiac arrest, wherein a patient arrives with chest pains ultimately requiring defibrillation and cardiopulmonary resuscitation. The second scenario involved pediatric anaphylaxis wherein a patient has a negative reaction to a medication. In this second scenario, an additional component of the simulation involves speaking with the parents of the child to inform them of the nature of the medical error leading to the anaphylaxis. We employed a zero-shot prompt approach but included two prompt conditions, a standardized text-based prompt, and a prompt that included an uploaded pdf flow chart that showed the LLM a template into how the scenario should unfold. We used a standardized prompt similar to the one used in our reference article (see supplementary materials). Regarding the prompt with flowchart, we had an additional prompting approach, wherein we uploaded a PDF attachment of a flow chart providing detail into an ideal scenario in which the simulation exercise should go. This flowchart was developed by our team. As these scenarios require specific adherence to medical procedures and

guidelines, we believe that it is important to determine if prompting that leverages upfront educator effort, in this instance the flowchart scenario, increases output quality and fidelity.

We then used select metrics from our reference article [5]; Medical Accuracy, Realism, Persona Consistency, Fidelity to Prompt, Relevancy, and Usability, each rated on a 5-point Likert scale.

The metric 'empathy' was excluded as both our scenarios prioritized medical expertise over interpersonal dynamics. Our 3 clinician team members then provided ratings for each of these scripts, and we computed an average score across this metric per rater, giving us one overall score per dialogue. Each rater was informed of the two prompting styles, but were blinded to the model each dialogue was generated from.

Analytic Strategy

We examined descriptive statistics for average rating by way of LLM across all scenarios as well as mean scores by prompting condition. Prior research using our metric has set a cut off indicating a high-quality output at 4.5 or higher, hence we are using 4.5 as a high-quality indication [5]. In addition, we conducted a Poisson regression model to analyze word count by way of prompting condition to examine if prompting strategy impacts dialogue length. Prior literature has also demonstrated that LLMs can provide lengthy dialogues in specific scenarios, which may not always reflect realistic interactions [5]. Therefore, our analysis was conducted to assess if structured guidance impacted overall dialogue length and conciseness.

To analyze the quality of prompts, we employed a 3 x 2 x 2 repeated measures ANOVA on scenario, rater, and prompt condition, with each LLM effectively acting as a participant. This is in contrast to typical approaches, which analyze outputs at the level of individual rater [1-2]. If LLMs are to be used as partners in generating content, this analytic approach allows us to examine how raters essentially grade each performer under different scenarios and instruction types.

Results

Descriptives

We report average ratings for each metric by Prompting Condition. Table 1 showcases mean and SD ratings across metric by LLM. Interestingly, no group level mean by model meets the 4.5 high quality benchmark as evaluated by our expert raters. See Figure 1 for a visualization of the means by metric across model.

Word Count by Prompt Type

A Poisson regression model revealed a significant effect of prompt type on word count. Relative to standard dialogue prompts ($M = 1453$, $SD = 762$), prompts that involved a detailed flow chart ($M = 1219$, $SD = 426$) generated shorter scripts across all models, $b = -0.18$, $p < .001$.

Effect of Prompt Type on Quality

Results revealed no significant main effects for rater $F(2,6) = 4.15$, $p = .07$, $\eta^2 = .32$, scenario $F(1,3) = 2.71$, $p = .2$, $\eta^2 = .00$, or prompt style $F(1,3) = 8.42$, $p = .06$, $\eta^2 = .08$ on overall output quality. There were also no significant interaction effects across the model including prompting style and scenario $F(1,3) = .655$, $p = .48$, $\eta^2 = .02$.

Discussion

Building off prior work, we prompted LLMs to generate dialogue for pediatric simulation cases, manipulating prompting style across each LLM and scenario. Results showed average scores did not meet previously defined thresholds for high quality outputs. While we applied similar evaluation criteria from previous work [5], these results suggest that the different context of our scenarios did not translate to high quality outputs. In addition, outputs generated from a standardized text-based prompt generated longer scripts by way of raw word count compared to a prompt with a more detailed flow chart attached. We then examined how each LLM was rated

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4 in terms of dialogue quality across both scenarios and prompting types, with results showing no
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6 significant differences in quality across our expert raters, scenarios, or prompting style.
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9 Our flow chart prompt involved more work up front from our research team where expertise
10 guided the LLM in how the scenario should unfold. Thus, in this condition, the LLMs did not
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12 need to solve the scenario, merely provide dialogue around the structure. This may explain the
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14 shorter dialogues relative to the ones using a standardized prompt. Surprisingly, no effect in
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16 quality was detected between these prompt styles. Educators aiming to use LLMs to create
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18 learning materials in the form of physician/team dialogues may benefit from structuring lessons
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20 around learner appraisal of such dialogues, wherein learners identify clinical/procedural
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22 inaccuracies and improvements to communication. This is in relation to our finding that mean
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24 scores do not always meet the benchmark of high-quality dialogues by the ratings rubric used in
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26 this study.
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32 Our approach used variations on zero shot prompting. Recent work has shown iterative
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34 prompting approaches can improve quality of LLM outputs [9]. If LLMs can be used as thought
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36 partners [10] in generating educational material, an iterative process of initial generation, review,
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38 and refinement with subsequent prompting may be a better overall strategy in the use of LLMs
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40 for material creation. Future research should further examine how providing more context such
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42 as worked examples of clinical activities as well as examples of high quality dialogue impact
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44 output quality.
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48 Our analytic approach differed from our reference article for expansion in that we analyzed our
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50 data at the level of LLM and not expert rater. Here, we treated the LLMs themselves as
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52 participants, akin to assessing how students would be graded on a series of tasks by different
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54 assessors. Work on LLM evaluation has primarily analyzed data at the level of a small to
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56 moderate size of expert raters [1-2, 11], where many measurements create enough variance to
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58 capture significant differences. Future research could employ alternative strategies to increase
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sample size. For example, a study where many outputs from one or multiple LLMs are generated but with one experimental condition, splitting the total number of outputs. With a series of raters, the data could then be analyzed at the level of output, with a small n of raters treated as a random effect [12], along with whatever experimental manipulation was conducted as well as LLM comparison. This would allow us to assess individual outputs and their quality as the main unit of analysis, potentially allowing us to make better inferences on LLM outputs.

Finally, work on LLMs has shown that models are increasingly being iterated, improved, and trained on with new developments growing rapidly [13]. This presents a challenge for slow methodical research on the outputs of LLM generation. More research is needed on identifying best practices for leveraging LLMs in the development of educational material regardless of advances in model outputs or performances.

Regarding limitations, the authors of the reference article [5] did not specify how they developed the high quality benchmark as 4.5 over 5, which should be a point of caution, and clarification of the benchmark could be an important next step.

Conclusions:

In conclusion, our findings suggest that average quality scores of the 4 LLM outputs did not meet previously defined thresholds for high quality, although they came close. Therefore, our work should guide clinicians and educators in choosing LLM and prompting strategies for generating educational content as well as research on evaluating LLM outputs.

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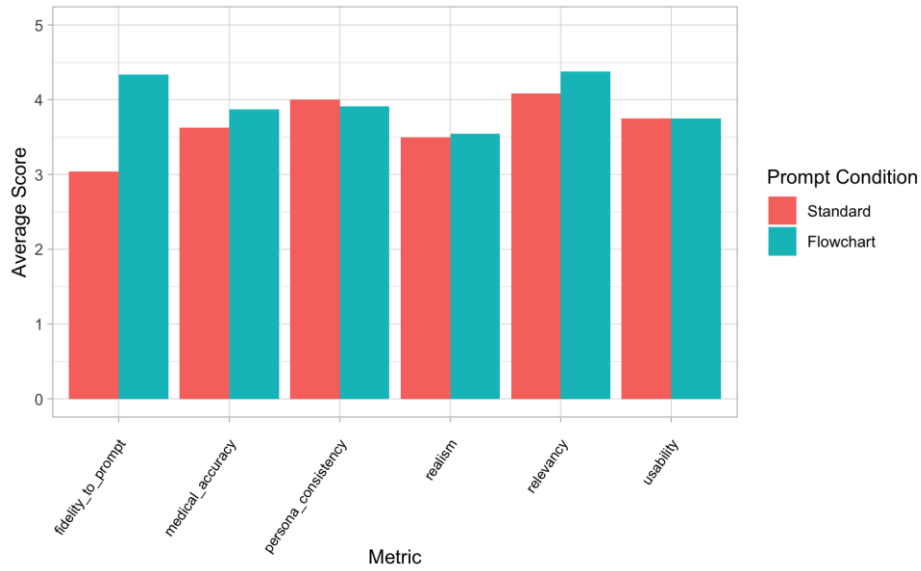


Figure 1: Plot of average scores by metric across Prompt Condition

Metric	Prompt Condition	Average Score	SD
Fidelity to Prompt	Dialogue Prompt	3.04	1.23
Medical Accuracy	Dialogue Prompt	3.62	1.35
Persona Consistency	Dialogue Prompt	4	0.59
Realism	Dialogue Prompt	3.5	1.06
Relevancy	Dialogue Prompt	4.08	0.72
Usability	Dialogue Prompt	3.75	0.85
Fidelity to Prompt	Flow Chart	4.33	0.87
Medical Accuracy	Flow Chart	3.88	1.03
Persona Consistency	Flow Chart	3.92	0.58
Realism	Flow Chart	3.54	0.88
Relevancy	Flow Chart	4.38	0.71
Usability	Flow Chart	3.75	0.74

Table 1: Mean and Standard Deviations of metric quality by Prompt Condition